

Bloodwork Cheat Sheet

BabyBird Physique | Wesnoski Coaching

How to read this sheet

Your lab's "normal" range tells you you're not sick. The functional range shows where your body tends to run best. Both columns are below for every marker.

Conventional range: what your lab report uses. It's built from the middle 95% of people who got tested, and many of them weren't healthy. Exact numbers vary by lab, so check your own report.

Functional range: the tighter window many functional practitioners aim for. Landing outside it doesn't mean something is wrong. It means it's worth a closer look.

Units: US units (mg/dL, ng/mL). If your lab reports in other units, the numbers will look different.

One number never tells the whole story. Patterns across markers, plus how you actually feel, are what matter.

Thyroid

TSH alone misses a lot. Ask for the full panel so you can see how much active hormone you're actually making.

Marker	Conventional range	Functional range	What it tells you
TSH	~0.45-4.5 mIU/L	1.0-2.0 mIU/L	The brain's signal to the thyroid. Higher means it's pushing harder to get output.
Free T4	~0.8-1.8 ng/dL	1.3-1.7 ng/dL	The storage form of thyroid hormone.
Free T3	~2.0-4.4 pg/mL	3.4-4.2 pg/mL	The active form. Drives energy, metabolism and body temperature.
Reverse T3	~9-25 ng/dL	Under 12 ng/dL	The "brake" form. Rises with long diets, stress and inflammation.
Free T3 ÷ Reverse T3	Not reported	0.2 or higher	How well you convert T4 into usable T3.
TPO antibodies	Below lab cutoff (varies)	Under 9 IU/mL	Immune activity against the thyroid. Positive = autoimmune flag.
Thyroglobulin antibodies	Below lab cutoff (varies)	Under 4 IU/mL	A second autoimmune marker. Can be positive when TPO is not.

Typical pattern: TSH "normal" around 3-4, low-normal free T3, cold hands, low energy, stalled fat loss.

Blood sugar and insulin

Fasting insulin often climbs years before blood sugar does. Most standard panels don't include it, so ask for it by name.

Marker	Conventional range	Functional range	What it tells you
Fasting glucose	~70-99 mg/dL	75-85 mg/dL	Blood sugar at the moment of the draw.
Fasting insulin	~2.6-24.9 µIU/mL	2-6 µIU/mL	How hard your body works to keep sugar in range. An early warning sign.
HbA1c	Under 5.7%	4.8-5.3%	Your average blood sugar over about 3 months.
HOMA-IR	Not reported (over ~2.5 often used as a cutoff)	Under 1.5	An insulin resistance score. $\text{Glucose} \times \text{insulin} \div 405$.

HOMA-IR is easy to calculate yourself: multiply fasting glucose (mg/dL) by fasting insulin, then divide by 405.

Inflammation and heart health

These show how much background inflammation you're carrying and how your body handles fats.

Marker	Conventional range	Functional range	What it tells you
hs-CRP	Under 3.0 mg/L (under 1 = low risk, 1-3 = average, over 3 = high)	Under 1.0 mg/L, ideally under 0.5	General inflammation. Hard training raises it for a day or two.
Homocysteine	Under ~15 µmol/L (varies by lab)	6-8 µmol/L	Linked to B vitamin status and blood vessel health.
Triglycerides	Under 150 mg/dL	60-100 mg/dL	Blood fats. Often high with too many refined carbs or insulin resistance.
HDL	40+ mg/dL (men), 50+ mg/dL (women)	Over 60 mg/dL (men), over 70 mg/dL (women)	The "cleanup" cholesterol.
Triglyceride ÷ HDL ratio	Not reported	Under 1.5, ideally near 1	A simple insulin resistance check. Over 3 is a red flag.
ApoB	Under ~90 mg/dL (varies by lab)	Under 80 mg/dL	The count of particles that can build plaque. Often more useful than LDL alone.

Iron and nutrients

Low-normal iron stores and vitamin levels are some of the most common reasons people feel tired with "normal" labs.

Marker	Conventional range	Functional range	What it tells you
Ferritin	~30-400 ng/mL (men), ~15-150 ng/mL (women)	100-150 ng/mL (men), 70-100 ng/mL (women)	Iron stores. Also rises with inflammation, so read it next to hs-CRP.
Iron saturation	~15-55%	25-35%	How much iron is actively moving in your blood. Over 45% is worth checking further.
Vitamin D (25-OH)	30-100 ng/mL	40-50 ng/mL	Affects immunity, mood, hormones and bone health.
Vitamin B12	~200-1,250 pg/mL	600-1,000 pg/mL	Energy, nerves and red blood cells. Recent supplements can make it read high.
Folate (serum)	Over ~3-5.4 ng/mL (varies by lab)	Over 15 ng/mL	Works with B12. Low folate can raise homocysteine.
RBC magnesium	~3.7-7.0 mg/dL (varies by lab)	5.5-6.5 mg/dL	Magnesium inside your cells. Better than serum magnesium, which stays "normal" until you're very low.

Hormones

Hormone ranges depend heavily on sex, age, the lab's test method and, for women, the day of your cycle. Use these as a guide and compare against your own report.

Marker	Conventional range	Functional range	What it tells you
Total testosterone (men)	~264-916 ng/dL	700-1,100 ng/dL	Drives muscle, drive, recovery and mood.
Total testosterone (women)	~2-60 ng/dL (varies by age and lab)	30-70 ng/dL	High can point to PCOS or insulin resistance. Low can mean low drive and poor recovery.
Free testosterone	Varies a lot by test method	Upper half of your lab's range	The usable portion that isn't bound up.
SHBG (men)	~10-56 nmol/L (higher after 50)	20-40 nmol/L	Binds testosterone. Low often goes with insulin resistance. High can lock up free T.
SHBG (women)	~17-125 nmol/L	40-70 nmol/L	Low is a common insulin resistance and PCOS flag.
Estradiol (men)	~8-43 pg/mL	10-30 pg/mL	Men need some estrogen for joints, libido and mood. Too high or too low both cause problems.
Estradiol (women)	Depends on cycle day	Read by cycle day	Test on cycle day 3 for a clean baseline unless your provider says otherwise.
DHEA-S	Depends on age and sex	400-600 µg/dL (men), 200-300 µg/dL (women)	An adrenal hormone. Low can show up with long-term stress.

Marker	Conventional range	Functional range	What it tells you
Morning cortisol (serum)	~6-19 µg/dL (varies by lab)	13-15 µg/dL	Your stress hormone at its daily peak. Draw between 7 and 9 AM.

A single cortisol draw is a snapshot. A 4-point saliva or dried urine test shows your full daily rhythm.

Liver and kidney (read these if you lift)

Hard training can make healthy lifters look like they have liver or kidney problems on paper. Know the difference before you panic.

Marker	Conventional range	Functional range	What it tells you
ALT	~7-56 U/L (varies by lab)	15-25 U/L	Liver enzyme. Muscle also releases it, so it rises after heavy training.
AST	~10-40 U/L	15-25 U/L	Found in liver and muscle. Often jumps more than ALT after a big lift day.
GGT	~0-60 U/L (varies by lab and sex)	10-25 U/L	Comes mostly from the liver, not muscle. Ask for it when ALT and AST are high.
Creatinine	~0.7-1.3 mg/dL (men), ~0.6-1.1 mg/dL (women)	Read alongside muscle mass	More muscle and creatine supplements both raise it, even with healthy kidneys.

Lifter rules of thumb:

High ALT/AST with normal GGT usually points to muscle, not liver. Retest after 3-5 days off lifting.

If creatinine looks high, ask about cystatin C. It checks kidney function without being skewed by muscle mass.

Before your draw

How you prepare can change your numbers as much as your health does.

- [] Fast 10-12 hours. Water is fine.
- [] Book a morning draw, ideally 7-9 AM.
- [] No hard training for 48 hours before. 3-5 days if you're checking liver enzymes.
- [] Stop biotin 48-72 hours before, or up to a week on high doses (over 5 mg). It can throw off thyroid and hormone results.
- [] Skip B12 and other supplements the morning of the draw.
- [] Women: note your cycle day. Day 3 is standard for hormone testing.
- [] Stay well hydrated the day before.
- [] Keep a copy of every result so you can compare over time.

Important

This sheet is for education only. It isn't medical advice and doesn't diagnose or treat anything. Always review your results with your physician, and don't change medications based on these ranges.

Want help reading your results?

Numbers only matter when someone connects them to how you feel and what you're doing. That's what the R3 System is built on: Regulate, Repair, Rebuild.

If your labs came back "normal" but you don't feel normal, book a quick call to see if coaching is the right fit.

Sources

Conventional ranges were checked against Labcorp reference intervals: reverse T3, free T3, insulin, testosterone, SHBG, estradiol, ferritin, homocysteine, RBC magnesium, vitamin B12, folate, ApoB and AM cortisol. GGT: UW Medicine. Biotin timing: SiPhox Health. Functional ranges reflect common functional-medicine practice and are not lab standards.

Work with Jimmy: <https://info.wesnoscicoaching.com/jimmy-application>